Syed Ahsan Masud Zaidi

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SUMMARY

- Ph.D. candidate in Computer Science focused on Computer Vision, Video Understanding, and Machine Learning.
- Developed video analysis methods for sports medicine and athlete safety using Python and PyTorch.
- Research experience in offline reinforcement learning and robotic control.
- Published author at **IROS**, **ICPR**, **CVPR** workshop (CVsports) and **RRPR**.

PROFESSIONAL EXPERIENCE

Data-Driven Discovery (DDD) Lab, Kansas State University 2023–Present

Doctoral Researcher | *Computer Vision and Video Understanding*

- Built GRAZE for zero-shot contact localization using Grounding DINO, SAM2, and motion analysis.
- Fine-tuned ViViT for tackle classification and released the TackleNet dataset and benchmark software.
- Benchmarked ViViT, MViTv2, SlowFast, and VideoMAE v2 with session-disjoint evaluation and leakage checks.
- Authored CRISP (IROS 2026) on masked skill inference for long-horizon offline robotic control.

University of Engineering and Technology Lahore 2014–2023

Lecturer | *Research and Laboratory Leadership*

- Directed the Robotics Lab, supporting applied work in mechatronics and control engineering.
- Provided technical guidance in programming, embedded systems, electronics, and control engineering.
- Managed semester planning and departmental coordination as In-Charge of the Semester Cell.

Black Gold Solutions (Plan9) 2014

Founder & CEO | *Startup* | *Fuel Management Technology*

- Developed a product for efficient fuel management.
- Led technical development, product strategy, and startup operations.
- Won third prize at the TiE Startup Challenge 2014.

PepsiCo International 2013–2014

Project Engineer (Automation and Commissioning)

- Installed and commissioned a KRONES PET filling plant, integrating equipment and automation systems.
- Diagnosed electrical and control-system faults during commissioning and production startup.
- Tested automated equipment and supported production handover and operational readiness.

TECHNICAL SKILLS AND SOFTWARE

Programming: Python, C, C++, MATLAB.
Machine learning: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers.
Computer vision: Action recognition, event localization, detection, segmentation, OpenCV, RetinaFace.
Foundation models: Grounding DINO, SAM2, SAM 3.1; zero-shot inference and fine-tuning.
Evaluation: Class-imbalance handling, cross-validation, session-disjoint evaluation, leakage checks.
Computing: Git, Linux, SLURM; GPU training on Beocat and PSC Bridges-2.
Robotics and engineering: ROS, control systems, signal processing, embedded systems, LabVIEW.

EDUCATION

Kansas State University 2023–Present

Ph.D. Candidate in Computer Science | GPA: **3.96/4.0**. Advisor: Lior Shamir.

Research: computer vision, video understanding, and AI for sports safety.

King Fahd University of Petroleum and Minerals 2019

M.S. in Electrical Engineering | Dhahran, Saudi Arabia.

Thesis research: laser-beam position control for free-space optical communication.

University of Engineering and Technology Lahore 2013

B.S. in Mechatronics and Control Engineering | Lahore, Pakistan.

Thesis: Static Marker-Based Cricket Bowling Analysis.

SELECTED PUBLICATIONS

1. **GRAZE: Grounded Refinement and Motion-Aware Zero-Shot Event Localization**
   Zaidi et al. | *IEEE/CVF CVPR Workshops, Computer Vision in Sports (CVSports), 2026. Spotlight. Paper*

2. **ViTs for Action Classification in Videos: An Approach to Risky Tackle Detection in American Football Practice Videos**
   Zaidi et al. | *International Conference on Pattern Recognition (ICPR), 2026. Paper | Publisher*

3. **CRISP: Context-Robust Skill Inference for Long-Horizon Offline Robotic Control**
   Zaidi et al. | *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. Code*

4. **Reproducing Vision Transformer-Based Risky Tackle Detection in American Football Practice Videos: A Companion Study**
   Zaidi et al. | *RRPR Workshop at ICPR, 2026. Oral presentation; Springer LNCS post-proceedings forthcoming. Conference listing*

OPEN SOURCE RESEARCH SOFTWARE

TackleNet benchmark resources Dataset | Code

Released a PyTorch dataset loader, fixed split files, baselines, and shared evaluation utilities for technique classification and contact localization. Used RetinaFace and SAM 3.1 for face de-identification.

GRAZE inference workflow Code

Provided inference scripts, environment setup, SLURM batch jobs, and prediction aggregation to support reproducible contact-localization experiments.

TECHNICAL LEADERSHIP AND PROFESSIONAL SERVICE

- **Research communication:** Presented a CVPR Workshops spotlight and poster, an ICPR main-conference poster, and an RRPR oral presentation in 2026.
- **Peer review:** IEEE Access; SLAS Technology; Reproducible Research in Pattern Recognition (RRPR) at ICPR; Women in Computer Vision (WiCV) at ECCV.

AWARDS AND RECOGNITION

- **AOSSM Playmaker Grant Lab team finalist**, American Orthopaedic Society for Sports Medicine, 2026.
- **CVPR Broadening Participation support**, Computer Vision Foundation, 2026.
- **IAPR Travel Stipend and Registration Scholarship**, ICPR 2026.
- **Full Merit Scholarship**, M.S. studies at KFUPM.